TRANSFORMER EXPLOSION & OIL-SPILL HAZARD SIMULATOR

Module 2 — Technical Documentation

Internal-arcing fault · tank overpressure · oil-spill reach · safe clearance · in-app calibration

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Version 1.5 · June 2026
Pressure chain calibrated & frozen · runtime Excel→JSON calibration

Abstract

This document specifies the model and methodology behind Module 2 of the *Proximity of Urban Networks* toolset: a standalone simulator that estimates the hazard produced when an internal arcing fault occurs inside an oil-filled distribution transformer. From four inputs — rated power, voltage, internal fault current and fault-clearing time — the tool estimates the peak tank overpressure, the maximum distance reached by ejected boiling oil, the resulting required safe clearance, and an overall hazard score.

As of **Version 1.5**, the full pressure chain is experimentally grounded on the full-scale internal-arcing tests of Yan et al. (IEEE Transactions on Power Delivery, 2023). The arc-energy formulation and the pressure-coupling coefficient are calibrated and frozen, reproducing the measured peak pressures of two distinct transformers (315 kVA / 35 kV and 40 MVA / 110 kV) to within a mean absolute error of 14 %. The methodology is further cross-checked against CIGRE Technical Brochure 620 (WG A3.24), whose composite energy-rate coefficient independently coincides with the value derived here.

Version 1.5 also introduces an **in-application calibration workflow**: measured tests supplied as an Excel or CSV file are fitted at the click of a button, written to a JSON coefficient file, and applied immediately on the next run — so the empirical coefficients can be re-derived from new evidence without editing the source code.

The wider motivation is that medium-voltage clearance rules should not be rigidly generalised; they are more defensible when adapted to local geography and climate. Module 2 contributes a **data-driven hazard quantity** — the oil-spill reach — to which such clearance decisions can be anchored. The estimator is deliberately transparent and lightweight rather than a full CFD solve, so that it remains understandable to a general working-group audience. To the best of our knowledge the formulation is physically consistent and, for the calibrated pressure chain, validated against published full-scale tests.

Contents

1. Introduction and Scope

The study on the proximity of urban networks is organised into two independent modules so that each can be developed, validated and presented on its own:

- **Module 1 — Overhead-line mechanical clearance:** conductor sag, wind pressure, conductor and ambient temperature, mechanical weight, and the resulting phase-to-phase and phase-to-ground clearances.
- **Module 2 — Transformer oil-spill hazard (this document):** tank overpressure, oil-spill throw distance, required safe clearance and an overall hazard score for an internal arcing fault.

1.1 Purpose

Module 2 turns four routine nameplate / protection quantities into a quantified spatial hazard. The intent is decision-support oriented: to make visible how the hazard footprint of a transformer changes with rating, voltage, available fault current and protection speed, and therefore why a single generalised clearance figure can be either over-conservative or unsafe depending on the installation.

1.2 Scope and assumptions

- Applies to oil-filled distribution transformers subject to an internal arcing fault; reference case 225 kVA / 13.8 kV.
- Produces engineering estimates for screening and comparison, not type-test or certification values.
- Resolves the peak (first-pulse) dynamic overpressure and a single representative spill trajectory; full dynamic pressure-wave propagation and tank fragmentation are not modelled.
- All empirical coefficients are exposed in one configuration block and may be re-calibrated locally — either by editing the block or, since v1.5, through the in-application calibration workflow (Section 5).

1.3 What is new in Version 1.5

This release supersedes the v1.3 documentation. The principal differences are summarised below; the remainder of the document describes the v1.5 model as built.

Area	v1.3 (previous)	v1.5 (this document)
Arc energy	Scaled from phase voltage $V\varphi \times I \times t \times k$	Arc-voltage gradient: $0.9 \times (G \times \text{gap}) \times I \times t$; validated ±4 %
Overpressure	Constant-volume gas heating (k_heat, γ); quasi-static	Calibrated peak $\Delta p = A\_E \cdot E / V\_oil$; frozen on full-scale tests
Calibration	Coefficients = unvalidated defaults	Pressure chain frozen (MAPE 14 %, R^2 0.93) + runtime Excel→JSON fitting
Out-of-domain	Not flagged	$E/V > 80$ kJ/m³ flagged as transient indicator; ejection capped at rupture
Interface	Light theme	Dark engineering dashboard; calibration bar; isometric hazard map

2. Physical Background

An internal arc in mineral oil rapidly vaporises and decomposes the surrounding liquid, generating a large volume of gas in a few milliseconds. Because the tank is effectively a closed vessel, this gas production drives a steep pressure rise. The first pressure pulse is dominated by the dynamic expansion of the arc-generated gas bubble; it can substantially exceed the value that a purely quasi-static (ideal-gas heating) estimate would predict. If the pressure-relief path is insufficient, the tank wall is stressed and may deform or rupture, ejecting hot and possibly burning oil into the surrounding area.

Full-scale experiments reported in the literature characterise this sequence — the oil-pressure rise and fluctuation, bubble pulsation, and tank deformation — and provide validated numerical (bubble–fluid–solid) models against which simplified estimators can be calibrated. Module 2 adopts a transparent reduced-order chain and treats the published full-scale data of Yan et al. as its calibration reference, with the CIGRE TB 620 switchgear-arc methodology as an independent cross-check of model structure.

2.1 Why the peak pressure is calibrated, not derived

A first-principles constant-volume estimate (the v1.3 approach) under-predicts the measured first peak by roughly an order of magnitude, because it omits the dynamic bubble transient. Rather than introduce a speculative dynamic model, v1.5 collapses the dynamics into a single measured coupling coefficient `A_E` relating the peak overpressure to the arc energy density. This coefficient is fitted to — and frozen against — full-scale measurements, and is the central empirical statement of the model.

3. Methodology

The model evaluates a short chain of physically-shaped expressions. Each empirical coefficient is named and listed in Section 4. The symbols are summarised below.

3.1 Nomenclature

Symbol	Quantity	Unit
E_arc	Arc energy injected into the oil	J
G_arc	Arc-voltage gradient in oil	V/cm
L_gap	Arc gap length (lead 5 cm / winding 1 cm)	cm
I_eff	Effective (bounded) RMS fault current	A
t	Fault-clearing time	s
S_n	Transformer rated power	kVA
S_sc	Network short-circuit capacity	MVA
I_sc	Available network short-circuit current	kA
Z_pu	Transformer short-circuit impedance	p.u.
V_oil	Tank oil inventory	m^3 (L)
E/V_oil	Arc energy density (calibration variable)	kJ/m^3
A_E	Pressure-coupling coefficient (calibrated)	$bar \cdot m^3/J$

Symbol	Quantity	Unit
Δp	Peak tank overpressure	bar
v_e	Oil ejection velocity	m/s
R	Maximum oil-spill distance	m
R_safe	Required safe clearance	m
ρ_oil	Mineral-oil density (880)	kg/m³

3.2 Fault-current bounding

Before any energy is computed, the entered fault current is bounded by the level physically available at the fault location. For an arc on the HV **lead / bushing** inside the tank — the scenario behind most violent tank ruptures — the current is fed directly by the upstream network and the transformer impedance does not limit it. The ceiling is the network short-circuit current, computed from the short-circuit capacity *S_sc* at the selected primary voltage:

$$I_{sc} = S_{sc} / (\sqrt{3} \cdot V_{LL}) \qquad \text{(0a)}$$

For a **winding** (through-type) fault, the transformer's own short-circuit impedance limits the current; the default *Z_pu* follows the IEC 60076-5 minimum-impedance ladder (4.0 % up to 630 kVA, rising to 12.5 % above 63 MVA). All downstream steps use the bounded effective current:

$$I_{eff} = \min(I_{entered} , \text{ceiling}) \qquad \text{(0b)}$$

Default network fault levels. The default short-circuit parameters follow the Egyptian Electricity Holding Company (EEHC) standard specifications: **11 kV** → 500 MVA (EEHC EDMS 08-400-01); **22 kV** → 500 MVA (EEHC EDMS 08-400-01 & 08-200-2).

Note: the 500 MVA value is the network short-circuit capacity (fault level), not the transformer rating. The simulator converts it to I_sc at the selected voltage (500 MVA → 26.2 kA at 11 kV; 20.9 kA at 13.8 kV; 13.1 kA at 22 kV). These are representative assumptions and may be replaced by utility-approved local values through the user-override input.

3.3 Arc energy (validated formulation)

The energy delivered to the oil is the product of the arc voltage, the effective current and the clearing time. The arc voltage is set by the **gap geometry**, not by the system voltage: full-scale measurements give an arc-voltage gradient of order 60–88 V/cm across systems from 35 kV to 110 kV. A waveform factor of 0.9 (= 2√2/π, the composite of a square-wave arc voltage and a sinusoidal current) converts peak to effective contribution:

$$E_{arc} = 0.9 \cdot (G_{arc} \cdot L_{gap}) \cdot I_{eff} \cdot t \qquad \text{(1)}$$

With G_arc = 73 V/cm and L_gap = 5 cm (lead) or 1 cm (winding). This formulation reproduces the measured arc energies of the calibration tests to within ±4 %, and its 0.9 factor independently coincides with the composite energy-rate coefficient of CIGRE TB 620 (eq. 3-1).

3.4 Decomposition gas (reporting only)

An indicative decomposition-gas volume is reported, proportional to the arc energy through a gas-yield coefficient $y = 8.5\times10^{-8}$ m³/J (85 cc/kJ at STP). Since v1.1 this is a reporting output only; the overpressure is computed directly from the arc energy.

$$V\_gas = y \cdot E\_arc \qquad (2)$$

3.5 Tank oil inventory

The oil volume available to absorb the pressure rise is scaled from the rating:

$$V\_oil = c\_oil \cdot S\_n \qquad (c\_oil = 1.2\ \text{L/kVA}) \qquad (3)$$

3.6 Peak overpressure (calibrated chain)

The central calibrated relation. The peak overpressure is proportional to the arc energy density E/V_oil through the coupling coefficient `A_E`:

$$\Delta p\ [\text{bar}] = A\_E \cdot E\_arc\ /\ V\_oil \qquad (A\_E = 4.08\times10^{-5}\ \text{bar}\cdot\text{m}^3/\text{J}) \qquad (4)$$

A_E is fitted by least squares through the origin on the full-scale measurements (Section 6) and **frozen** as “Calibration v1”. The relation is validated for E/V_oil up to ≈ 80 kJ/m³ (the tested range). Beyond that, the result is reported as a **transient indicator** rather than a physical tank pressure, and the interface flags the case as extrapolated.

3.7 Oil ejection velocity

The ejection velocity at the rupture / relief path follows Torricelli’s relation for an overpressure driving an incompressible liquid, with discharge coefficient C_d = 0.70. Because the tank ruptures and relieves near 2 bar, the driving pressure is capped at that value: out-of-domain pressures are transient indicators, not sustained tank states.

$$\Delta p\_eject = \min(\Delta p,\ 2.0\ \text{bar})\ ; \quad v\_e = C\_d \cdot \sqrt{2 \cdot \Delta p\_eject\ /\ \rho\_oil} \qquad (5)$$

3.8 Maximum spill distance and safe clearance

The maximum throw is the ballistic range at the optimal 45° launch, reduced by an efficiency factor $\eta = 0.483$ that lumps drag, fragmentation and non-ideal launch. A safety factor of 1.5 sets the required clearance:

$$R = \eta \cdot v\_e^2\ /\ g\ ; \qquad R\_safe = 1.5 \cdot R \qquad (6)$$

Calibration status: the reach block (η·C_d²) remains the one un-validated link, pending full-scale spill-distance data (e.g. SERGI / Transformer Protector). The v1.5 calibration workflow already accepts a measured-reach column and will fit this block automatically once such data is supplied (Section 5.4).

3.9 Hazard score and protection zone

A 0–100 hazard score combines the overpressure and the oil reach, each normalised against a reference level and equally weighted:

$$Risk = 100 \cdot (\ 0.5 \cdot \Delta p/1.70 + 0.5 \cdot R/9.0\ )\ , \text{ clamped to } [0,100] \qquad (7)$$

The protection zone is assigned from the peak overpressure: CONTROLLED below 0.5 bar (a sound pressure-relief device is expected to cope), MODERATE up to 2.0 bar, and RUPTURE RISK at or above 2.0 bar.

4. Model Coefficients

All empirical coefficients live in a single configuration block. The pressure-chain values are frozen as Calibration v1; the reach values are provisional pending validation.

Coefficient	Symbol	Value	Status / Source
Arc-voltage gradient	G_arc	`73 V/cm`	Frozen — Yan 2022 measured (59.5–87.5)
Arc gap (lead / winding)	L_gap	`5 / 1 cm`	Frozen — geometry nominal
Waveform factor	—	`0.9`	Frozen — 2√2/π; matches CIGRE TB 620
Pressure coupling	A_E	**4.08×10^{-5}**	FROZEN — fitted on Yan 2022 (R^2 0.93)
Valid energy density	—	≤ 80 kJ/m^3	Frozen — tested range
Gas yield	y	8.5×10^{-8}	Literature — 85 cc/kJ STP (reporting)
Oil per kVA	c_oil	`1.2 L/kVA`	Engineering default
Discharge coefficient	C_d	`0.70`	Provisional — reach block
Ejection cap	—	`2.0 bar`	Frozen — tank rupture threshold
Throw efficiency	η	`0.483`	Provisional — reach block
Safety factor	—	`1.5`	Engineering default
Oil density	ρ_oil	880 kg/m^3	Mineral oil

5. In-Application Calibration Workflow

Version 1.5 turns calibration from a code edit into an in-tool operation. Measured tests are supplied as a spreadsheet, fitted with one click, written to a JSON coefficient file, and applied on the next run. The frozen defaults of Section 4 remain the fall back whenever no calibration file is present.

5.1 Operating sequence

1. **Browse…** — select a measured-tests file (.xlsx or .csv).
2. **Calibrate** — the tool reads the rows, fits the coefficient(s), writes `calibration_active.json` beside the program, and reports the fit (A_E, R^2, MAPE, pass/fail against the 15 % criterion, the worst point, and the valid domain).
3. **Run** — every run re-reads the JSON, so the new coefficients apply **immediately**, with no source edit. Deleting the JSON restores the frozen defaults.

5.2 Input file schema

Column headers are matched case- and space-insensitively, with common aliases accepted. A row needs a measured pressure and an oil volume, plus either a measured arc energy or the current and duration (from which the energy is estimated via Eq. 1).

Column	Meaning	Required
Transformer_Rating_kVA	Rated power	context

Column	Meaning	Required
Voltage_kV	Primary voltage	context
Fault_Current_kA	Test current (used if energy absent)	conditional
Arc_Duration_ms	Arc duration (used if energy absent)	conditional
arc_energy_kj	Measured arc energy (preferred)	preferred
Oil_Volume_Liters	Tank oil inventory	**yes**
Measured_Pressure_Bar	Measured peak overpressure	**yes**
Measured_Reach_m	Measured spill distance (enables reach fit)	optional
Source_Reference / Test_ID	Label for the per-point report	optional

5.3 Generated JSON

The calibration file records the fitted coefficient and full quality metrics, including a per-point table of measured vs predicted pressure and percentage error, the count of measured vs estimated-energy points, and the auto-derived valid domain. It is human-readable and may be archived alongside the test campaign as a calibration record. The `generated` timestamp and `source_file` name make each calibration traceable.

5.4 Reach calibration (forward-compatible)

When a `Measured_Reach_m` column is present in two or more rows, the same Calibrate action also fits the reach coefficient $B = \eta \cdot C_d^2$ (against the capped driving pressure) and reports its MAPE against the 20 % criterion. This is the mechanism by which the outstanding reach block will be validated once full-scale spill data is available, with no code change required.

5.5 Worked calibration on the supplied template

Running the workflow on the four-row template (three Yan 2022 rows plus one synthetic laboratory row without measured energy) returns $A_E = 5.70\times10^{-5}$ with $R^2 = 0.19$ and MAPE = 33 % — a deliberate **FAIL**. The per-point report localises the cause: the synthetic row predicts 2.5 bar against a stated 5.2 bar (−52 %), i.e. it is inconsistent with the Yan trend. Removing that row recovers $A_E = 4.08\times10^{-5}$ and $R^2 = 0.93$ — exactly the frozen default. The example demonstrates that the tool both fits and **audits** incoming data: an inconsistent test is surfaced rather than silently absorbed.

6. Validation of the Pressure Chain

The frozen coefficient A_E was fitted on three full-scale anchor points spanning two transformers and a 6× range of energy density. Predicted peak pressures agree with measurement to a mean absolute error of 14 %, satisfying the project acceptance criterio n (< 15 %).

Test (source)	Rating / Voltage	E/V_{oil} (kJ/m^3)	Measured (bar)	Predicted (bar)	Error
Yan 2022 — Test 1	315 kVA / 35 kV	38.1	1.96	2.17	+10.6 %
Yan 2022 — Test 2	315 kVA / 35 kV	75.9	2.88	2.97	+3.3 %
Yan 2022 — Test 3	40 MVA / 110 kV	10.1	0.48	0.48	+0.5 %

Independent cross-check (CIGRE TB 620). The brochure's composite energy-rate coefficient ($2\sqrt{2}/\pi = 0.9003$) coincides with the waveform factor derived here; its arc-voltage description is likewise geometry-based rather than system-voltage based; and its stated accuracy class (≈ ±10 % with measured energy, larger predictively) matches the 14 % obtained here. The structural agreement strengthens confidence in the model form independently of the Yan fit.

6.1 Limitations

Item	Status in v1.5
Pressure chain	Calibrated & frozen (MAPE 14 %, R^2 0.93)
Reach block ($\eta \cdot C\_d^2$)	Provisional — awaiting full-scale spill data; calibratable in-app
Energy density > 80 kJ/m³	Extrapolated — reported as transient indicator, flagged in UI
Mechanical rupture / deformation	Not modelled; driving pressure capped at 2 bar
Multi-phase arc dynamics	Single representative arc via I_eff
Temperature-dependent oil properties	Constant properties; acceptable below ≈150 °C

7. Worked Examples

Three representative runs illustrate the dominant levers. The reference case is deliberately at the edge of the calibrated domain to show the extrapolation flag in action.

Case	E_arc (kJ)	E/V (kJ/m³)	Δp (bar)	Reach (m)	Safe (m)	Zone
225 kVA / 13.8 kV / 5 kA / 100 ms (lead)	164.2	608	24.82 ⚠	10.97	16.45	RUPTURE
225 kVA / 13.8 kV / 5 kA / 10 ms (lead)	16.4	61	2.48	10.97	16.45	RUPTURE
50 kVA / 11 kV / 5 kA / 100 ms (winding)	0.4	7	0.29	1.61	2.41	CONTROLLE

⚠ = energy density beyond the calibrated domain; the 24.82 bar value is a transient indicator, not a sustained tank pressure. The middle row (identical except a 10× faster clearing time) brings the case back into domain and drops the indicated pressure ten-fold, illustrating that **clearing time is the dominant safety lever**. The winding case shows the model's physical consistency at the small-arc end: a 1 cm gap yields a tiny arc energy and a CONTROLLED outcome.

8. Software and Interface

Module 2 is a single self-contained Python file (standard library plus tkinter; openpyxl optional, only for reading .xlsx calibration files). It runs with one command and is published as a single repository asset.

8.1 Layout

- **Input bar:** rating, voltage, fault current, duration, fault location, network S_sc, and an optional I_sc override.

- **Calibration bar:** Browse / Calibrate controls and a live badge showing whether the active coefficients are the frozen default or a custom JSON (with its A_E, R^2 and point count).
- **Status strip:** the effective current, its ceiling and basis, and an amber warning when the case is extrapolated.
- **KPI cards:** peak pressure (amber-tagged “transient indicator” when extrapolated), oil reach, safe distance, and a colour-coded protection zone.
- **Risk assessment:** a 0–100 multi-colour gauge and a per-factor breakdown.
- **Hazard map:** an isometric plate with three glowing zone rings (rupture core, oil reach, safe clearance) around a schematic transformer; resize-aware.
- **Recommendation and Summary:** safety actions (including extrapolation guidance) and a parameter recap that names the active calibration and its R^2.

8.2 Reproducibility

Because every run re-reads `calibration_active.json`, a result is fully described by the four inputs plus the named calibration record. Archiving the JSON with a study therefore fixes the coefficients used, supporting reproducible analyses across the working group.

9. Standards and Regulatory Context

- **IEC 60076-5** — transformer ability to withstand short circuit; basis for the winding-fault impedance ladder.
- **CIGRE TB 620 (WG A3.24)** — internal-arc pressure-rise simulation in switchgear; independent methodological cross-check and source of the relief/burst design logic informing the roadmap.
- **EEHC EDMS 08-400-01 / 08-200-2** — Egyptian utility network short-circuit levels; basis for the default fault ceilings.

The tool produces screening estimates for clearance and siting decisions and is not a substitute for type testing or certification.

10. Roadmap

1. **Reach calibration:** ingest full-scale spill-distance data and fit $\eta \cdot C_d^2$ through the existing calibration workflow (v1.5.1).
2. **Rupture-disk / relief designer (v1.5+):** size a relief opening so the peak pressure equals the disk’s response pressure independently of tank volume, following the CIGRE TB 620 overshoot criterion — the engineering core of the “< 1 MVA protection concept”.
3. **Documentation maintenance:** keep this report synchronised with each frozen calibration and with the validated reach block when available.

References

[1] Yan, C., et al. “Research on Oil Pressure Rise and Fluctuation Due to Arcing Faults Inside Transformers.” IEEE Transactions on Power Delivery, 38(2), 1483–1496, 2023.

[2] CIGRE Working Group A3.24. Tools for the Simulation of Pressure Rise Due to Internal Arc in MV and HV Switchgear. Technical Brochure 620, CIGRE, 2014.

[3] IEC 60076-5. Power Transformers — Part 5: Ability to Withstand Short Circuit. International Electrotechnical Commission.

[4] Egyptian Electricity Holding Company (EEHC). Distribution Network Standard Specifications, EDMS 08-400-01 and EDMS 08-200-2.